



Kiran Peddi

Senior Platform & Site Reliability Engineer | AWS/EKS | IaC, Observability & Developer Platforms
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Visa: U.S. Permanent Resident (Green Card holder - no sponsorship required) **actively pursuing U.S. citizenship naturalization**

Security Clearance: Public Trust eligible; eligible for U.S. federal programs and government employment.

PROFESSIONAL SUMMARY

- Senior Site Reliability and DevOps Engineer with 9+ years designing and operating highly available AWS infrastructure and distributed microservices in regulated financial services, healthcare, and telecommunications environments. Experienced in Agile delivery teams, SLO/error-budget practices, incident command, and automated release controls.
- Deep operational expertise defining Service Level Indicators (SLIs) and Service Level Objectives (SLOs) in CloudWatch, Prometheus, and Grafana; experienced in enforcing error budget policies and release gates to balance engineering velocity with system stability.
- Proven Incident Commander for major production outages, leading real-time war room coordination, structured Root Cause Analysis (RCA), blameless post-mortems, and preventive engineering remediation in ServiceNow.
- Advanced expertise in architecting multi-AZ resilient cloud topologies and provisioning immutable AWS environments using AWS CloudFormation templates and HashiCorp Terraform modules with strict state management, CI/CD validation, and policy-as-code linting.
- Extensive background implementing zero-downtime deployment strategies (blue/green, canary) integrated with ServiceNow ITSM change workflows, CAB approval gates, and ITIL release standards in strict SOC 2, HIPAA, and financial audit environments.
- Proficient in developing reusable automation packages, CLI tooling, and API integrations in Python and Go, enforcing software engineering rigor through unit testing, code reviews, semantic versioning, and CI validation.
- Experienced in containerizing microservices and managing multi-node Amazon EKS environments supporting more than 500 production pods, including autoscaling, resource limits, ingress routing, and service-mesh components.
- Adept at troubleshooting distributed Java (Spring Boot), Node.js, and React applications running on Apache Tomcat and Netty web servers, diagnosing memory leaks, JVM thread contention, and connection pool bottlenecks.

CORE TECHNICAL SKILLS

- **Cloud Infrastructure:** Amazon Web Services (AWS - EC2, EKS, S3, RDS PostgreSQL/MySQL, Lambda, VPC, IAM, Route53, CloudFront), GCP, Azure
- **Infrastructure as Code (IaC):** AWS CloudFormation, HashiCorp Terraform, Red Hat Ansible Automation
- **Observability & Telemetry:** CloudWatch Metrics/Logs, Prometheus, Grafana, Datadog, Distributed Tracing, AWS X-Ray



- **ITSM & Incident Management:** ServiceNow (Incident, Change, Problem Management), ITIL v4 Frameworks, PagerDuty
- **Containers & Orchestration:** Kubernetes (EKS), Docker, Helm, Container Registries (ECR)
- **CI/CD & Delivery Pipelines:** Jenkins, AWS CodePipeline/CodeBuild, GitLab CI/CD, GitHub Actions, Automated Rollback Strategies
- **Languages & Scripting:** Python, Go, Bash / Shell Scripting, SQL
- **Application Stack & Web Servers:** Java / Spring Boot, Node.js, React, Apache Tomcat, Netty, REST APIs, Microservices
- **Databases & Storage:** AWS RDS (PostgreSQL, MySQL), DynamoDB, Redis Caching, Connection Pooling / ORM Drivers
- **Security & Governance:** SOC 2, ISO 27001, HIPAA, AWS Security Hub, DevSecOps, IAM Least Privilege

PROFESSIONAL EXPERIENCE

Senior Site Reliability Engineer

May 2023 – Present

Mayo Clinic | Rochester, MN

Stack: AWS (EC2, EKS, RDS, S3, IAM, CloudWatch), CloudFormation, Terraform, Spring Boot, Node.js, Python, Prometheus, Grafana, ServiceNow, PagerDuty, ISO 27001 / HIPAA

- Partnered with product, application, security, and platform teams in Scrum delivery cycles to prioritize reliability backlog items, SLO remediation, and release-readiness work.
- Re-architected critical clinical data ingestion services for high availability and disaster recovery across multi-AZ AWS environments, introducing automated Route53 health-checked failover, cross-region RDS read replicas, and dynamic pod auto-scaling to maintain 99.99% system resilience during regional network degradation.
- Led reliability engineering for 15+ critical healthcare services, defining p99 latency and availability SLOs in Prometheus/Grafana; standardized alert thresholds and routing for 40+ stakeholders, reducing false-positive pages by 25% based on PagerDuty alert analytics data.
- Served as Incident Commander for 12+ Severity 1 production incidents, coordinating cross-functional engineering war rooms, completing blameless RCAs within 3 business days, and driving corrective action plans recorded in ServiceNow incident records that reduced incident recurrence by 30%.
- Co-designed error-budget policies for 20+ production services with 99.9% availability targets; integrated Prometheus burn-rate signals with GitLab release gates to pause high-risk deployments and require documented risk acceptance for exceptions.
- Designed and maintained reusable CloudFormation and Terraform modules across 12+ AWS accounts and environments, embedding security scanning, syntax linting, and policy validation in CI/CD before production changes.
- Built a Python CLI for automated Amazon EKS upgrades and secret rotation, with unit tests, semantic versioning, and CI validation; adopted by three platform teams and eliminated approximately 15 manual operational hours per month.



- Integrated ServiceNow ITSM change management APIs directly into GitLab CI/CD pipelines to automate risk assessments, compliance audit logs, and CAB approval gates for production deployments.
- Diagnosed JVM thread contention and database connection-pool exhaustion across Spring Boot and Node.js services running on EKS and Tomcat; reduced p99 latency by 35% through thread-pool tuning and established connection-pool sizing standards for future services.
- Correlated CloudWatch logs, Prometheus metrics, and AWS X-Ray traces to isolate latency regressions in Spring Boot services and prioritize remediation.
- Authored operational runbooks, disaster recovery (DR) failover procedures, and incident response playbooks; conducted weekly code reviews for 6 junior SREs on infrastructure troubleshooting and on-call operational practices.

Cloud Infrastructure & DevOps Engineer

September 2021 – May 2023

Clario | Philadelphia, PA

Stack: AWS (EC2, S3, RDS PostgreSQL, CloudWatch, VPC), CloudFormation, Jenkins, Python, Bash, Tomcat, ITIL/ServiceNow, SOC 2

- Built Jenkins and Python delivery pipelines for 10+ Spring Boot services, automating unit testing, container vulnerability scanning, blue/green deployments, and health-checked rollbacks; reduced manual release steps by 60% and increased deployment frequency from bi-weekly to on-demand based on Jenkins deployment logs.
- Constructed AWS CloudFormation stacks automating multi-region VPC networking, subnets, NAT gateways, and secure IAM roles supporting clinical trial data processing engines compliant with SOC 2 requirements.
- Implemented CloudWatch metrics, centralized logs, and service alerting for 25+ environments, detecting database-latency and HTTP 5xx regressions; reduced mean time to detect (MTTD) from 45 minutes to under 5 minutes based on CloudWatch alarm logs.
- Coordinated with Change Advisory Boards (CAB) to conduct risk assessments for high-impact releases, implementing automated health checks and automatic deployment rollback scripts in Jenkins.
- Conducted root cause investigations on recurring infrastructure alerts, rewriting Ansible playbooks to resolve disk space bloat and log rotation failures across EC2 fleets.

Cloud Infrastructure Engineer

February 2019 – August 2021

Comcast | West Chester, PA

Stack: AWS, CloudFormation, Ansible, Python, Tomcat, Netty, React, Linux, ServiceNow

- Developed Python and Shell automation for database-snapshot retention and OS patch verification across 150+ EC2 hosts, eliminating 25+ manual operational hours per month and maintaining 99.5% patch-compliance reporting based on internal security audit records.
- Managed AWS compute and web server infrastructure for high-throughput customer portals utilizing AWS CloudFormation templates and Ansible playbooks.



- Configured and tuned Apache Tomcat and Netty application servers, optimizing memory allocation, thread pool sizing, and reverse-proxy routing to handle high traffic spikes.
- Acted as Tier-3 escalation engineer for infrastructure outages, analyzing system logs, netstat metrics, and stack traces to isolate and remediate network and middleware bottlenecks.

DevOps Engineer / AWS Engineer
JPMorgan Chase | Westerville, OH

March 2018 – January 2019

Stack: AWS, Jenkins, Docker, Bash, Python, SQL, Financial Services Security Frameworks

- Standardized Jenkins pipelines for Dockerized microservices, adding automated static security scanning, artifact signature verification, and immutable deployment gates for 15+ repositories supporting sensitive financial transactional data systems.
- Enforced strict financial-services release controls, segregation of duties (SoD), and audit trail logging across production deployment workflows to meet enterprise risk and regulatory compliance requirements.
- Collaborated with cybersecurity and risk teams to enforce IAM least-privilege policies, security group boundary constraints, and KMS envelope encryption for core transactional databases.
- Executed automated database backup and migration routines across AWS RDS instances using custom Python scripts and AWS CLI tasks.

IT Infrastructure Specialist / Build & Release Engineer
FedEx | Orlando, FL

March 2017 – February 2018

Stack: Linux, Shell Scripting, Jenkins, Apache Tomcat, Oracle DB, ITIL

- Executed build and release management across pre-production and production environments following strict ITIL release control and change ticket procedures in enterprise ITSM systems.
- Automated manual build steps using Shell scripting and Jenkins jobs, reducing human error in software releases.

CERTIFICATIONS

- **Cloud & Infrastructure Architecture:** AWS Certified Solutions Architect – Associate
- **Kubernetes & Automation:** HashiCorp Certified Terraform Associate | Red Hat Certified Ansible Automation Specialist
- **Cloud Security & Networking:** Google Cloud Certified Professional Cloud Security Engineer | Google Cloud Cybersecurity Specialist | DevSecOps & Google Security Operations Fundamentals | Google Cloud Advanced Networking (Architecture, Routing)
- **Databases & Security:** Cybersecurity Specialist Professional Certification | SQL and Relational Databases 101 (Cognitive Class)
- **Secondary Continuous Professional Development:** Vertex AI Prompt Design & Generative AI Architecture | Large Language Models & Responsible AI Engineering Frameworks

EDUCATION

- **Master of Science in Electrical and Electronics Engineering** **August 2014 - December 2016**
 University of Hartford, West Hartford, CT, U.S.



- **Bachelor of Science in Electronics and Communication Engineering** June 2009 - May 2013
Jawaharlal Nehru Technological University, India.